Addition and subtraction of algebra[redacted]ctions

1 Simplify:

a $\dfrac{2x}{7}+\dfrac{7x}{7}$ b $\dfrac{3x}{4}+\dfrac{2x}{4}$ c $\dfrac{2x}{3}+\dfrac{10x}{3}$ d $\dfrac{3x}{12}+\dfrac{12x}{12}$

e $\dfrac{-8x}{9}+\dfrac{4x}{9}$ f $\dfrac{-3x}{14}+\dfrac{-2x}{14}$ g $\dfrac{8x}{2}-\dfrac{3x}{2}$ h $\dfrac{13x}{3}-\dfrac{8x}{3}$

i $\dfrac{7x}{5}-\dfrac{10x}{5}$ j $\dfrac{31x}{6}-\dfrac{7x}{6}$ k $\dfrac{25x}{24}-\dfrac{5x}{24}$ l $-\dfrac{4x}{7}-\dfrac{5x}{7}$

2 Simplify:

a $\dfrac{2x}{3}+\dfrac{x+3}{3}$ b $\dfrac{7x}{8}+\dfrac{5x+2}{8}$

c $\dfrac{-2x}{7}+\dfrac{4x-3}{7}$ d $\dfrac{5x-1}{3}-\dfrac{2x+3}{3}$

e $\dfrac{4x-10}{3}+\dfrac{-5x+10}{3}$ f $\dfrac{-6x-8}{5}+\dfrac{-3x-5}{5}$

g $\dfrac{4x^2+11}{6}-\dfrac{3x^2-4x}{6}$ h $\dfrac{2x^2-3}{5}+\dfrac{3x^2+6}{5}$

i $\dfrac{-5x^2-2x}{3}+\dfrac{-3x^2-2x}{3}$ j $\dfrac{2x-7x^3}{4}-\dfrac{-x^2-3x^3+1}{4}$

3 Find the lowest common denominator for the following pairs of algebraic fractions:

a $\dfrac{w}{2}$ and $\dfrac{w}{6}$ b $\dfrac{s}{3}$ and $\dfrac{s}{6}$ c $\dfrac{x}{2}$ and $\dfrac{y}{3}$ d $\dfrac{r}{5}$ and $\dfrac{s}{3}$

4 Consider the algebraic fractions $\dfrac{2m}{5}$ and $\dfrac{3m}{6}$.

a Find the lowest common denominator. b Hence, simplify $\dfrac{2m}{5}+\dfrac{3m}{6}$.

5 Complete the following:

$$\frac{3x}{5}+\frac{2x}{\square}=\frac{\square}{35}$$

6 Simplify:

a $\dfrac{2x}{5}+\dfrac{4x}{7}$ b $\dfrac{2x}{14}+\dfrac{14x}{21}$ c $\dfrac{4x}{5}-\dfrac{x}{7}$ d $\dfrac{4x}{45}-\dfrac{x}{18}$

e $-9x-\dfrac{5x}{7}$ f $\dfrac{3x}{5}-\dfrac{2x}{7}$ g $-3x-\dfrac{3x}{8}$ h $\dfrac{11x}{14}+\dfrac{7x}{21}$

i $\dfrac{x}{15}+\dfrac{3x}{35}$ j $\dfrac{4x^2}{3}-\dfrac{5x^2}{2}$

7 Simplify:

a $\dfrac{2x}{3}+\dfrac{2x+5}{9}$

b $\dfrac{8x}{13}+\dfrac{3x+4}{39}$

c $\dfrac{3x}{10}-\dfrac{4x+3}{16}$

d $\dfrac{4x}{9}+\dfrac{2x-3}{36}$

e $\dfrac{8x}{11}-\dfrac{2x-3}{44}$

f $\dfrac{y}{10}-\dfrac{5y+1}{25}$

g $\dfrac{7x}{8}-\dfrac{5x+6}{20}$

h $\dfrac{x+3}{3}+\dfrac{3x-2}{12}$

i $\dfrac{x+3}{3}-\dfrac{5x-7}{12}$

j $\dfrac{2x+3}{4}-\dfrac{2x+2}{12}$

k $\dfrac{5x+8}{6}+\dfrac{x+2}{2}$

l $\dfrac{3x+4}{6}+\dfrac{3x+4}{5}$

m $\dfrac{2x+2}{4}-\dfrac{x+3}{8}$

n $\dfrac{6x+7}{3}-\dfrac{3x-5}{9}$

o $\dfrac{6x^2}{7}+\dfrac{x^2+5x}{21}$

p $\dfrac{2x^2}{7}+\dfrac{3x^2+2x}{28}$

8 Simplify:

a $\dfrac{3}{x}+\dfrac{5}{y}$

b $\dfrac{c}{ab}+\dfrac{2}{bc^2}$

c $\dfrac{2x^2+4y}{y}-\dfrac{x-3}{x}$

d $\dfrac{4}{x}+\dfrac{2}{x-2}$